Map of the Ladder

The Golden Ladder · opening

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The Golden Ladder: Golden Cuts from Plato to the Fourth Dimension · opening · 5 November 2026 · Licence CC BY-SA 4.0
DICTO proposed an opening and a closing chapter to frame the book, so that the rung chapters can go in between. Drafting with Claude (Anthropic).

Status (draft for DICTO's reading; this box is deleted when the opening goes into the volume)

- The order of the chapters is proposed: “Silence Is Golden” stays Chapter 2 as titled; the rung chapters follow as Chapters 3 to 8 in the order of Chapter 1's ladder table; the closing chapter is Chapter 9. DICTO's to confirm.
- Chapters 3 to 8 are DICTO's published papers, to be set as chapters word for word (as in The Golden Cycle and Star Cores); the next version of the book only after DICTO's read ok of the whole.

> *This book is a ladder. Each rung is a dimension, and on each rung something is golden: a bond in the plane, two cuts on every solid, a pair of four-dimensional solids, and at the top a pair of 600-cells inside E8. You can climb it in order, or open it at any rung; each chapter stands on its own and names the rung below it.*

The rungs

chapter	rung	title	paper
1	—	One Triangle, Three Numbers: the triangle of mirrors and its number 1, √2 or φ	opening chapter
2	3D–4D	Silence Is Golden: the ladder of multiplets	chapter
3	2D	The truncated triangle and the golden bond	#20, #48
4	3D	Every Platonic Solid Has Exactly Two Golden Cuts	#88
5	4D	The Red Queen 120-cell: the 120-cell atlas	#65
6	4D	All Five Golden Pairs Are Slices of Four-Dimensional Solids	#89
7	6D	Golden on the Octahedron, Silver on the Cube	#90
8	4D	Four-Dimensional Solids Whose Every Cell Is a Golden Cut	#91
9	8D	The Top Rung: E8 and its golden twist	closing chapter

How to read it

Chapter 1 gives the one idea everything else uses: the five Platonic solids come from three triangles of mirrors, and each triangle carries a single number, 1, √2 or φ. Wherever φ appears in a later chapter, it comes from the φ triangle. Chapter 2 hears the same numbers as music: the golden stations of the sweep from octahedron to cube are where ladders of particle multiplets converge. Chapters 3 to 8 climb from the plane to six dimensions, one golden result per rung, each a paper of DICTO's series set as a chapter. Chapter 9 stands at the top: in eight dimensions E8 folds into two 600-cells in golden ratio, and the twist that swaps them is the golden twist met on every rung.

Each chapter keeps its own figures, theorems and sources; the DICTO DOI of every paper is given at its chapter's head.